

Selectime: Validation-selected scopes and frozen forecast mixtures

1 Forecast family

Given an observed target prefix and horizon H , form frozen-backbone median forecasts from univariate input, native multivariate input, and retrieved covariates for $K \in \{1, 5, 10, 15, 20\}$. Chronos-2 supports every input; TS-ICL omits multivariate input and scope controls; Chronos-Bolt reports only univariate vanilla. Raw top- K forecasts are mixture inputs, not reported standalone methods.

2 Aligned retrieval

For retrieval lookback r , represent query or neighbor history x as

$$z(x) = \frac{x - \mu(x)}{\max(\sigma(x), 10^{-8})},$$

using finite lookback values only. Rank eligible neighbors by Euclidean distance on at least 80% finite overlap. Require a complete maximum- K list; smaller K values reuse ordered prefixes. The default datastore spans all items and variates in one dataset/frequency task; same-series scope remains an ablation. Task-specific calendar phases and strides define eligible dates.

Before supplying neighbor trajectory s_i , transform it to query units:

$$\tilde{s}_i = (s_i - \mu_i) \frac{\max(\sigma_q, 10^{-8})}{\max(\sigma_i, 10^{-8})} + \mu_q,$$

where both statistics use past retrieval lookbacks only. Supply the transformed neighbor histories and their already observed futures as covariates.

3 Validation chronology and hard scope selection

Validation origins use the official test phase and step backward by H before the first test origin. By default, their count is capped to the official test date count. Validation retrieval ends before validation starts; test retrieval ends before test starts. No test observation enters either datastore.

Score forecasts with MSSE: mean squared forecast error divided by the seasonal squared-difference scale from the observed prefix. The hard scope selectors compare only univariate and multivariate. The task selector averages variates within each validation date; the per-variate selector treats each item/variate separately. Adapttime’s paired moving-date-block bootstrap admits a scope when its mean loss difference from the observed best is at most one bootstrap standard error, preferring univariate within that threshold.

4 Win-frequency mixtures

The scope mixtures combine univariate and multivariate once per task or per item/variate. Each top- K mixture combines univariate with its corresponding retrieved-covariate forecast. For an alternative, exclude fallback and non-finite validation rows. Let n be paired usable rows and w alternative wins, with ties contributing one half. The frozen Beta(1,1) weight is

$$p = \frac{1 + w}{2 + n}, \quad \hat{y} = (1 - p)\hat{y}_{\text{uni}} + p\hat{y}_{\text{alt}}.$$

No usable row gives $p = 0$.

5 Closed-form scope ridge

The task-level scope ridge uses exactly the same validation references. Let $d = \hat{y}_{\text{multi}} - \hat{y}_{\text{uni}}$ and $r = y - \hat{y}_{\text{uni}}$. For finite target cells j , let a_j be the reciprocal of that row’s observed-future count times its seasonal scale. With fixed $\alpha = 1$, fit

$$p = \frac{\sum_j a_j d_j r_j}{\alpha + \sum_j a_j d_j^2}, \quad \hat{y} = \hat{y}_{\text{uni}} + p d.$$

The coefficient is not clipped. With no usable training row, record an unfitted fallback and use $p = 0$.

6 Support and evidence

A validation row requires finite support in its available context and future. Unusable rows never reach a backbone. Heuristic controls default to univariate when no validation evidence remains; scope ridge explicitly records its unfitted vanilla fallback. Official test support and the Seasonal Naive metric grid are unchanged. Earlier standalone-candidate and horizon-mixture evidence in the results recap is historical and does not define the current method set.